

Supplementary material

Gentle by knowing: how much must a care robot know about a person to lift their arm safely?

Mu-Hua (Maurice) Wang

National Yang Ming Chiao Tung University, Taiwan

This supplement accompanies the Brief Research Report. Tables S1–S3 list every preregistered hypothesis, the deviations from the protocol and the simulation parameters; Tables S4 and S5 give the counts behind main-text Figures 3 and 4; Table S6 lists the confirmation rounds; Figures S1 and S2 show the post hoc harm-threshold sweep and the post hoc trade-off analysis.

Differences are paired per person: estimate [95% CI], harmful-outcome or safe-completion rate, or mean harm in N·m for rounds 17–18. “Harmful” is ≥ 0.5 N·m unless marked 2.5. Source files are the evaluation logs `out/round*/eval*.txt` and `out/audit_claims.txt` in the repository.

Table S1. Every preregistered hypothesis and its outcome

ID	Hypothesis	Result
H17-1	Population body model (routing, no declining), record error 8°, lower mean harm than F ($n = 40$)	not supported: +0.57 [−1.04, +2.18]
H17-2	... lower than reactive guard R	not supported: +0.01 [−1.88, +1.89]
H17-3	Tolerance curve over record error 0/4/8/16°	monotone, every CI includes 0
H17-4	Restricted persons only	not supported: +1.03 [−1.91, +3.98]
H17-5	Gap to oracle	significant: +1.77 [+0.24, +3.30]
H18-1	Probe-identified body model (4°) lower mean harm than F ($n = 40$)	supported: −3.80 [−6.16, −1.45]
H18-2	... lower than R	not supported: −2.52 [−5.95, +0.91]
H18-3	Identification adds value over population model	supported (borderline): −1.89 [−3.77, −0.01]
H18-4	Record-error curve 0/4/8°	only 4° significant; non-monotone
H18R-1	Frozen replication of H18-1 ($n = 120$)	not supported: −0.94 [−2.16, +0.27]; completions 90 vs 107
H18R-2	Replication of H18-2	not supported: +1.19 [−0.74, +3.12]
H18R-3	Replication of H18-3	not supported: −0.10 [−0.95, +0.75]
H18R-4	Record-error curve	no significant error level
H19-1	Probing body model + declining: fewer harmful than F, safe completions within 3 ($n = 120$)	supported: −0.100 [−0.163, −0.037]; safe 67 vs 70
H19-2	... vs R	not supported: harmful −0.108 [−0.173, −0.044] but safe 67 vs 76
H19-3	Lower mean harm than F and R	supported: −3.96 [−6.82, −1.10]; −3.38 [−5.54, −1.23]
H19-4	Probing adds value over population model	not supported, reversed: harmful +0.067 [+0.013, +0.121] (29 vs 21)
H21-1	Body model vs record threshold at matched cost (3.5°)	inconclusive: −0.05 [−0.12, +0.02]; vs 2°: −0.08 [−0.16, −0.01]; vs 10.5°: +0.15 [+0.08, +0.22]
H21-2	Routing without declining no better than F (null)	supported: −0.03 [−0.09, +0.02]
H21-3	No force guard reduces harm by significantly > 5 points (hard)	supported: −0.03/−0.02/−0.01/+0.00
H21-4	Within-person Spearman(wrist force, load) < 0.5	supported: median 0.35
H22-1	H21-3 under 6° and 12° end-feel	supported (tightest guard at 6°: −0.04 [−0.08, −0.00])
H22-2	Body model + declining vs F under 6°, 12°	supported: −0.21, −0.20
H22-3	Spearman < 0.5 under end-feel	6°: supported (0.30); 12°: not supported (0.75)
H22-4	Body model vs record threshold, $n = 600$, 4° records	model better: −0.06 [−0.09, −0.02]
H23-1	No force guard reduces harm by significantly > 5 points at 20°, 25° (0.5 and 2.5)	supported in all four cells; tightest guard −0.04 to −0.07 at half the completions
H23-2	Body model vs F at 20°, 25°	20°: supported (0.5, 2.5); 25°: supported at 0.5, not at 2.5 (safe 71 vs 76)
H23-3	Body model vs F with 10°, 20° record error	supported at 0.5; not at 2.5 (safe 387/394 vs 423)

ID	Hypothesis	Result
H23-4	Body model vs threshold equivalent (± 0.05) with 10–20° error	equivalent
H23-5	Categorical flag rungs (descriptive)	complete flag: 253 safe / 2 harmful; half: 303 / 102 ($n = 600$)
H24-1	L(20) fewer harmful than M(20), safe non-inferior (2 pp)	supported at 0.5: -0.14 [$-0.19, -0.09$]; not at 2.5 (safe -0.00 [$-0.04, +0.04$])
H24-2	L(10) vs M(10)	supported at 0.5: -0.14 [$-0.19, -0.10$]; not at 2.5
H24-3	L(20) vs F: fewer harmful, safe non-inferior (2.5 pp)	not supported: harmful -0.28 [$-0.34, -0.23$], safe -0.01 [$-0.05, +0.03$]
H24-4	Gap closed to oracle (descriptive)	63–69% (harm), 11–16% (safe)
H25-1	Replication: L vs M and L vs F	vs F supported (-0.27); vs M: harm -0.08 [$-0.12, -0.03$] but safe non-inferiority not met
H25-2	Half-missing records: L vs F	supported: -0.15 [$-0.19, -0.10$]
H25-3	Posture noise (SD 5°) / no posture: L vs M	supported: $-0.09, -0.07$
H25-4	Shifted population: L vs F and M	supported: $-0.53, -0.16$
H25-5	20° end-feel: L vs F and M	supported: $-0.33, -0.13$
A1	Fixed rule is not a straw man	supported: best hindsight fixed 203 harmful vs L 53
A2	No load shift to glenohumeral shear	supported: paired mean -3.6 N, median 0
A3	L robust to +10° stale records (harm rise $\leq 50\%$)	supported: $53 \rightarrow 68$ (M: $118 \rightarrow 210$)
H26-1	Reflex $\times 3$ / $\sigma = 0.40$ worlds: L vs F	supported: $-0.27, -0.31$
H26-2	... L vs M	supported: $-0.11, -0.17$

Table S2. Deviations from the protocol and reuse of persons

Item	Description
Bug, round 17	Joint ranges not restored between episodes in the robot’s internal body copy; found at the first round-17 evaluation; body-model arm re-run on the same 40 persons after F and R had been seen; affected output kept for audit.
Harm definition, round 22	Resting passive tension counted as harm under wide end-feel; found after round 22; net harm from round 23. Round-22 6° and 12° results use gross harm; 12° re-run under net harm in round 23 with the same conclusions; 4/120 persons affected at 12°.
Primary outcome	Paired mean harm (rounds 17–18) replaced by harmful-outcome rate plus safe-completion non-inferiority (round 19+).
Record error, round 18	Primary record error changed from 8° to 4° after round 17 had been seen.
Reuse of persons	Round-19 decline rule developed on rounds 18 + 18R (160); S threshold on rounds 18, 18R, 19 (280); L trained on rounds 18, 18R, 19, 21 (400). Round-21 hypotheses motivated by post hoc analyses of round 19. All later confirmation sets new.
Development only	Round 20 hybrid (body model + force guard + retries) did not improve on development data; not confirmed.
Analysis code	Round-25 set-precedence bug fixed during smoke test, before confirmation data. Confidence intervals use $t = 2.1$ for all $n > 20$ (conservative).
Post hoc	Threshold sweep; paired lifted-only comparisons; flags-only policy, type lookup, re-tuned M and S, trade-off curves.

Table S3. Simulation parameters

Parameter	Value	Source file
Simulator	MuJoCo 3.6.0; MyoSuite 2.12.2 myoArm, humerus free	<code>care_env6.py</code>
Stretch reflex	velocity gain 0.6×2 , length gain 4.0×2 , delay 25 ms, activation lag 30 ms, tonic 0.01	<code>reflex.py</code> , <code>care_env6.py</code>
Body variation	log-normal: strength, passive force, damping $\sigma = 0.25$; masses, inertias 0.7σ	<code>care_env.py</code>
End effector	capsule, 3 slide + 3 hinge servos, 6000 N/m, 20 N·m/rad, 50 Hz	<code>care_env6.py</code>
Task	rise 5 cm on observed forearm height, hold 0.5 s, max 3 s	<code>shoulder_task.py</code>
Strategies	8 sites (-6 to $+15$ cm, 3 cm), angles -40° / -20° / 0° , speeds 0.10/0.25 m/s	<code>round17.py</code>
Population	shoulder $p = 0.4$, $U(4, 16)^\circ$; elbow $p = 0.3$, $U(0, 10)^\circ$; shifted: 0.6, $U(2, 12)^\circ$; 0.5, $U(0, 8)^\circ$	<code>round17.py</code> , <code>round25.py</code>
Harm	peak 50-ms mean joint-limit torque (shoulder elevation + elbow flexion), first 100 ms excluded; net of pre-contact torque from round 23	<code>shoulder_task.py</code> , <code>dev_r24_learned.py</code>
Records	Gaussian error SD 4/10/20°, clipped at 0; half-documented $p = 0.5$	<code>round18.py</code> , <code>round25.py</code>
F	completion $\geq 90\%$ of 60 persons, lowest median harm	<code>round17.py</code>
R	$m \in \{0.6, 0.8, 1.0, 1.2\} \times p95$ harmless force; retract 0.5 s; ≥ 9 cm switch; 3 attempts	<code>round21.py</code>
M	199-body library; margin 3° ; predicted completion ≥ 0.8 ; decline if overshoot $> 3^\circ$	<code>round18.py</code> , <code>round21.py</code>
S	decline if recorded free range $< 3.5^\circ$	<code>round21.py</code>
L	kNN, $K = 25$, 6 standardised features, $\tau = 0.65$ (no-posture variant: 4 features, $\tau = 0.60$)	<code>dev_r24_learned.py</code> , <code>round25.py</code>
Misspecified worlds	reflex gain scale 2.0 $\rightarrow$ 6.0; body variation 0.25 $\rightarrow$ 0.40	<code>round26.py</code>

Table S4. Information ladder

Counts behind main-text Figure 3: 300 new persons, harmful ≥ 0.5 N·m (round 24).

Knowledge of the person	Safe completions	Harmful	Declined
None (fixed rule F)	177	110	0
Limitation flag, half documented $\rightarrow$ decline	140	60	88
Limitation flag, complete $\rightarrow$ decline	112	0	179
Ranges $\pm 20^\circ$ + record threshold S	155	51	83
Ranges $\pm 20^\circ$ + population body model M	162	67	59
Ranges $\pm 20^\circ$ + learned policy L	174	25	91
Ranges $\pm 4^\circ$ + M / L	176 / 184	64 / 20	49 / 85
Perfect knowledge O	237	0	63

Table S5. The learned policy under degradations

Counts behind main-text Figure 4: safe completions / harmful outcomes of F, M and L under degradations (harmful ≥ 0.5 N·m). “–”: not reported in this table; posture conditions were compared with M only, as preregistered (main-text Figure 4).

Condition (new persons)	F	M	L
Replication ($n = 300$)	171 / 108	170 / 51	167 / 28
Half of restrictions unrecorded	171 / 108	170 / 81	171 / 64
Posture noise, SD 5°	–	–	171 / 24
No posture input	–	–	174 / 29
Graded end-feel 20° ($n = 150$)	64 / 72	69 / 41	73 / 22
More, tighter contractures ($n = 250$)	72 / 170	70 / 77	77 / 38
Test physics: reflex gain $\times 3$ ($n = 300$)	168 / 108	152 / 61	158 / 28
Test physics: body variation $\sigma = 0.40$ ($n = 300$)	153 / 119	144 / 77	152 / 27

Table S6. Confirmation rounds

Within rounds 22 and 23, the end-feel conditions were run on the first 120 of the round’s 600 persons. Rounds 17 and 18 used paired mean harm as the primary outcome; rounds 19 onward used the harmful-outcome rate with a non-inferiority condition on safe completions.

Round	Question	New persons
17	Population body model choosing the support site (no declining) vs. fixed rule and reactive guard	40
18, 18R	Identifying the individual body by probing; frozen replication	40; 120
19	Body model with declining vs. fixed rule and reactive guard; value of probing	120
21	Force guard at four thresholds; routing without declining; model vs. record threshold	120
22	Graded end-feel (6°, 12°); model vs. record threshold	600
23	Wide end-feel (12–25°); record quality (error, flags, missing entries)	600
24	Information ladder including the learned policy L	300
25	Replication of L; missing records; posture noise; shifted population; 20° end-feel	300; 250
Audit	Strength of F; load shifting; stale records (reuses rounds 24–25)	(600)
26	Test physics differing from training: reflex gain $\times 3$; body variation $\sigma = 0.40$	300; 300

Figure S1. Sensitivity to the harm threshold (post hoc)

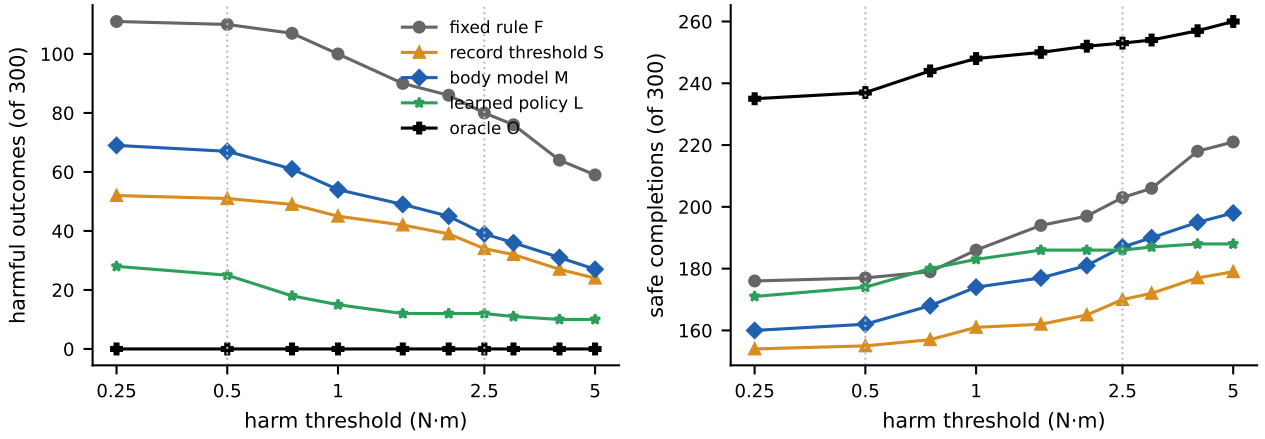


Figure S1. Post hoc sensitivity to the harm threshold (round 24, 300 new persons, record error 20°). Decisions are fixed; only the threshold used to classify outcomes varies. Dotted lines: the preregistered primary (0.5 N·m) and secondary (2.5 N·m) thresholds. Among F, S, M and L, L had the fewest harmful outcomes at every threshold (the complete-flag rule, which declines every flagged person, had none). The price of L's caution grew at lenient thresholds: F completed more lifts safely than L at almost every threshold, and by 8–33 lifts from 1.5 N·m upward (e.g. 203 vs. 186 at 2.5 N·m), while harming many more persons (80 vs. 12).

Figure S2. Trade-off between safe completions and harmful outcomes (post hoc)

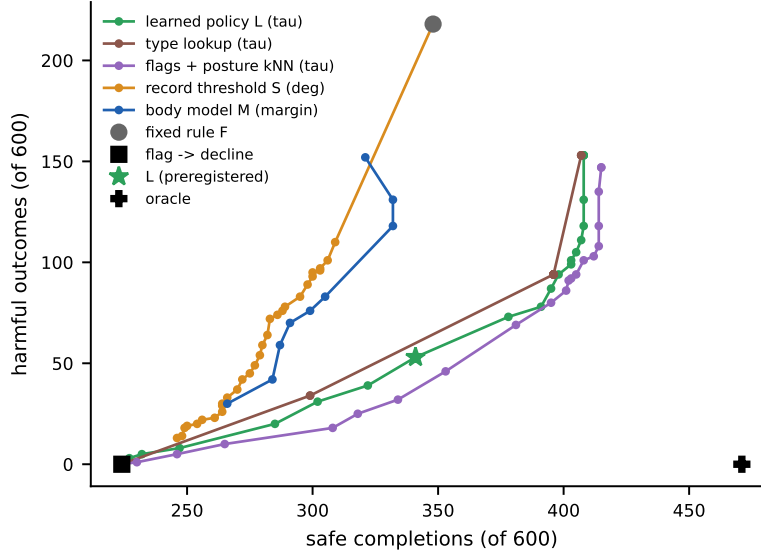


Figure S2. Post hoc trade-off between safe completions and harmful outcomes on the 600 confirmation persons with complete grids (rounds 24 and 25 A), sweeping each rule's decline parameter. Lower right is better. The learned policies (with or without recorded degrees) lie below the record-threshold and body-model curves. Baselines were tuned on the 400 training persons only, with the criterion used for τ . Re-tuned for 20° records, M (margin and decline threshold 6°) harmed 113 persons at 335 safe completions; the record threshold S could not meet the completion criterion at any setting and reverted to never declining. The type lookup (best training strategy per restriction type: none, shoulder, elbow, both) has only four distinct operating points; at the tuned setting it completed 396 lifts safely at 94 harmful outcomes. By restriction type, L lifted every unrestricted person with F (224 safe, 0 harmful), declined everyone with both joints restricted (71, of whom only 17 had any safe strategy), and chose between support sites and declining for single-joint restrictions, where it reduced harmful outcomes from 75 to 30 (shoulder) and from 82 to 23 (elbow) relative to F.